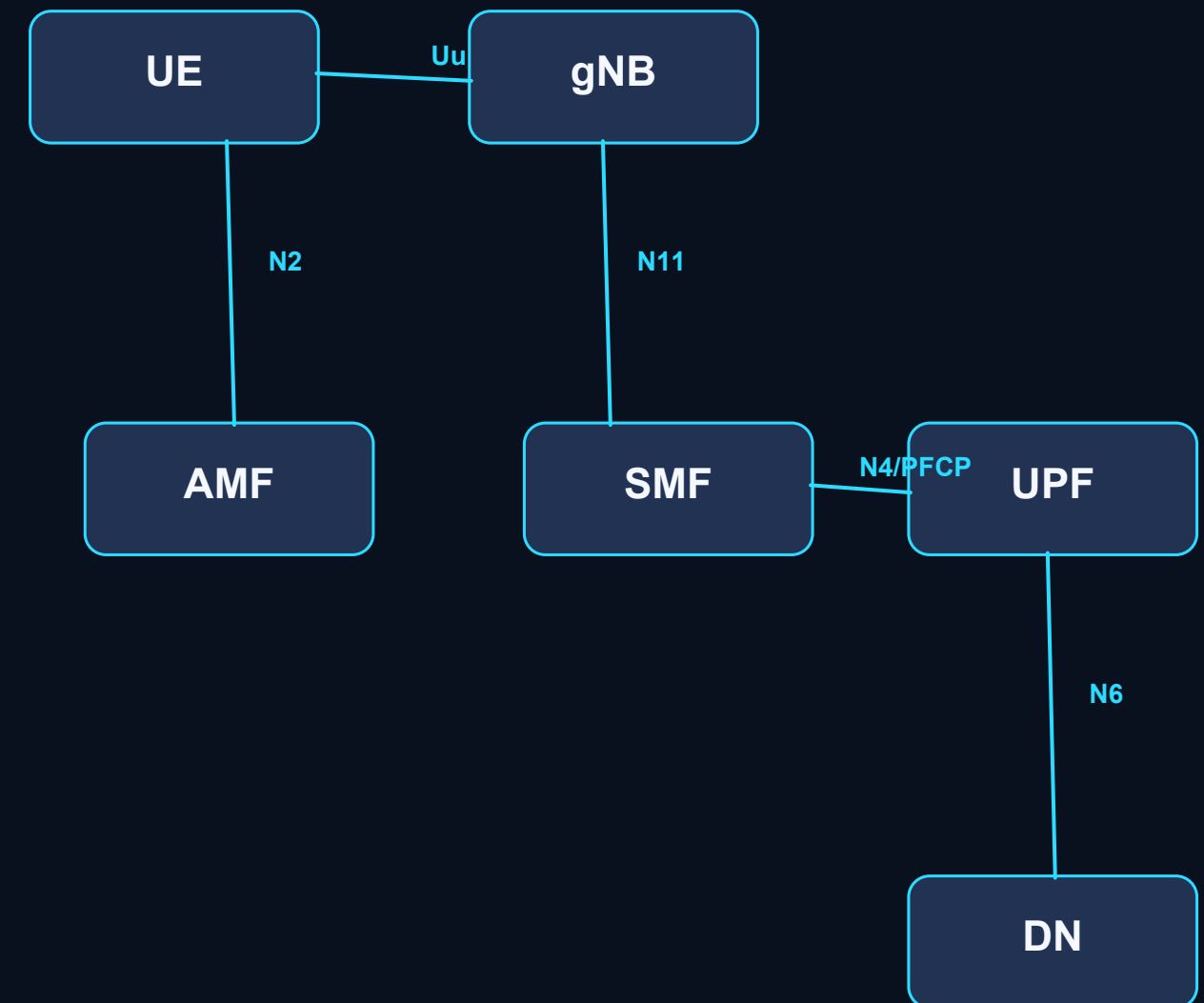


Simulation d'un réseau 5G et analyse d'une attaque PFCP

Déploiement avec UERANSIM et Open5GS

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Plan de la présentation

Une progression du déploiement vers l'analyse de sécurité PFCP

- 1 **Contexte et objectif du projet**
- 2 **Architecture de la simulation 5G**
- 3 Interfaces et protocoles utilisés
- 4 Déploiement de UERANSIM
- 5 Déploiement du cœur 5G Open5GS
- 6 Création de l'abonné
- 7 Configuration de l'UPF
- 8 Validation du réseau 5G
- 9 **Focus sur PFCP et analyse de l'attaque**
- 10 **Conclusion et recommandations**

Objectif du projet

Simuler un cœur 5G fonctionnel puis analyser une vulnérabilité protocolaire

Objectifs

Déployer une architecture 5G simplifiée

Connecter un UE simulé à Open5GS

Valider une session PDU fonctionnelle

Observer PFCP sur l'interface N4

Analyser une attaque PFCP

Outils

UERANSIM

Open5GS

MongoDB / WebUI

Wireshark

tcpdump

Idée centrale

PFCP permet au SMF de contrôler l'UPF.

Si cette interface est perturbée, le service data du UE peut devenir indisponible ou instable.

Architecture de la simulation 5G

plan utilisateur

Configuration du réseau des VMs

Configuration du réseau des VMs

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Custom	-	-	Enabled	192.168.200.0
VMnet1	Host-only	-	Connected	Enabled	192.168.233.0
LAN SiteA	Host-only	-	Connected	-	192.168.1.0

Ueransim

Power on this virtual machine
Edit virtual machine settings

▼ Devices

Memory

7.5 GB

Processors

4

Hard Disk (SCSI)

40 GB

CD/DVD (SATA)

Using file C:\User...

Network Adapter

NAT

Network Adapter 2

Custom (VMnet0)

USB Controller

Present

Sound Card

Auto detect

Display

Auto detect

open5gscp

Power on this virtual machine
Edit virtual machine settings

▼ Devices

Memory

7.5 GB

Processors

4

Hard Disk (SCSI)

32 GB

CD/DVD (SATA)

Using file C:\User...

Network Adapter

NAT

Network Adapter 2

Custom (VMnet0)

USB Controller

Present

Sound Card

Auto detect

Display

Auto detect

Open5gsup

Power on this virtual machine
Edit virtual machine settings

▼ Devices

Memory

7.5 GB

Processors

4

Hard Disk (SCSI)

40 GB

CD/DVD (SATA)

Using file C:\User...

Network Adapter

NAT

Network Adapter 2

Custom (VMnet0)

USB Controller

Present

Sound Card

Auto detect

Display

Auto detect

Communications principales

Communication	Protocole	Destination
UE vers gNB	RRC et NAS simulés	gNB <u>UERANSIM</u>
gNB vers AMF	NGAP sur SCTP 38412	<u>Open5GS</u> CP
gNB vers UPF	GTP-U sur UDP 2152	<u>Open5GS</u> UPF

Rôles des machines

UERANSIM : simule le UE et le gNB

Open5GS CP : héberge AMF, SMF, NRF, AUSF, UDM...

Open5GS UPF : transporte les données utilisateur

N4/PCF : lien critique entre SMF et UPF

Interfaces et protocoles utilisés

Chaque interface assure une fonction précise dans le réseau 5G

Interface	Communication	Protocole	Rôle
N2	gNB ↔ AMF	NGAP / SCTP 38412	Signalisation
N3	gNB ↔ UPF	GTP-U / UDP 2152	Données utilisateur
N4	SMF ↔ UPF	PFCP / UDP 8805	Contrôle des sessions
N6	UPF ↔ DN	IP	Accès réseau data

Point clé : l'interface N4 est la zone critique car elle transporte PFCP entre le SMF et l'UPF.

Installation et configuration de UERANSIM

Simulation du terminal utilisateur et de la station de base 5G

Installation UERANSIM

Partie 1 - Installation et configuration d'UERANSIM

Installer les paquets nécessaires à la compilation :

BUT Réseaux & Télécommunications

```
Shell
sudo apt update && sudo apt install -y cmake make gcc g++
libsctp-dev lksctp-tools iproute2 git
```

open5gs-gnb.yaml

```
Shell
sudo nano /opt/UERANSIM/config/open5gs-gnb.yaml
```

Configurer les paramètres principaux :

```
GNU nano 8.4 /opt/UERANSIM/config/open5gs-gnb.yaml
mcc: '999' # Mobile Country Code value
mnc: '70' # Mobile Network Code value (2 or 3 digits)
nci: '0x000000010' # NR Cell Identity (36-bit)
idlength: 32 # NR gNB ID length in bits [22...32]
tac: 1 # Tracking Area Code

linkIp: 192.168.200.130 # gNB's local IP address for Radio Link Simulation (Usually same with local IP)
ngapIp: 192.168.200.130 # gNB's local IP address for N2 Interface (Usually same with local IP)
gtpIp: 192.168.200.130 # gNB's local IP address for N3 Interface (Usually same with local IP)

# List of AMF address information
amfConfigs:
  - address: 192.168.200.128
    port: 38412

# List of supported S-NSSAIs by this gNB
slices:
  - sst: 1

# Indicates whether or not SCTP stream number errors should be ignored.
ignoreStreamIds: true

# Cell access type. When set to one of the satellite types (nr-leo, nr-meo,
# nr-geo, nr-othersat), the gNB attaches the NR-NTN TAI Information extension
# to every UserLocationInformationNR it sends to the AMF. Defaults to "nr".
cellAccessType: nr
```

open5gs-ue.yaml

```
GNU nano 8.4 /opt/UERANSIM/config/open5gs-ue.yaml
# IMSI number of the UE. IMSI = [MCC][MNC][MSISDN] (In total 15 digits)
supi: 'imsi-999700000000001'
# Mobile Country Code value of HPLMN
mcc: '999'
# Mobile Network Code value of HPLMN (2 or 3 digits)
mnc: '70'
# SUCI Protection Scheme : 0 for Null-scheme, 1 for Profile A and 2 for Profile B
protectionScheme: 0
# Home Network Public Key for protecting with SUCI Profile A
homeNetworkPublicKey: '5a8d38864820197c3394b92613b20b91633cbd897119273bf8e4a6f4eec0a650'
# Home Network Public Key ID for protecting with SUCI Profile A
homeNetworkPublicKeyId: 1
# Routing Indicator
routingIndicator: '0000'

# Permanent subscription key
key: '465B5CE8B199B49FAA5F0A2EE238A6BC'
# Operator code (OP or OPC) of the UE
op: 'E8ED289DEBA952E4283B54E88E6183CA'
# This value specifies the OP type and it can be either 'OP' or 'OPC'
opType: 'OPC'
# Authentication Management Field (AMF) value
amf: '8000'
# IMEI number of the device. It is used if no SUPI is provided
imei: '356938035643803'
# IMEISV number of the device. It is used if no SUPI and IMEI is provided
imeiSv: '4370816125816151'

# Network mask used for the UE's TUN interface to define the subnet size
tunNetmask: '255.255.255.0'

# List of gNB IP addresses for Radio Link Simulation
gnbSearchList:
  - 192.168.200.130
```

Paramètres cohérents entre UE, gNB et cœur 5G

MCC = 999 • MNC = 70 • TAC = 1 • AMF = 192.168.200.128 • gNB = 192.168.200.130 • Slice SST = 1

Déploiement du plan de contrôle Open5GS

Fonctions réseau chargées de l'enregistrement, de l'authentification et des sessions

Dépendances Open5GS

2. Installer les dépendances nécessaires à Open5GS

Installer les outils de compilation et les bibliothèques requises :

```
Shell
sudo apt update

Shell
sudo apt install -y python3-pip python3-setuptools python3-wheel \
    ninja-build build-essential flex bison git cmake libscrt-dev \
    libgnutls28-dev libgcrypt-dev libssl-dev libidn11-dev \
    libmongoc-dev \
    libbson-dev libyaml-dev libnghttp2-dev libmicrohttpd-dev \
    libcurl4-gnutls-dev libtins-dev libtalloc-dev meson
```

Exécutables Open5GS

Les exécutables utilisés pour le plan de contrôle doivent notamment être présents :

```
open5gscp@open5gscp:~/open5gs-2.7.7$ ls ~/open5gs-2.7.7/install/bin/
open5gs-amfd  open5gs-hssd  open5gs-nssfd  open5gs-scpd  open5gs-sgwud  open5gs-udrd
open5gs-ausfd  open5gs-mmed  open5gs-pcfd  open5gs-seppd  open5gs-smfd  open5gs-upfd
open5gs-bsfd  open5gs-nrfd  open5gs-pcrfd  open5gs-sgwcd  open5gs-udmd
open5gscp@open5gscp:~/open5gs-2.7.7$
```

nrf.yaml

```
GNU nano 8.4 nrf.yaml
logger:
  file:
    path: /home/open5gscp/open5gs-2.7.7/install/var/log/open5gs/nrf.log
  # level: info # fatal|error|warn|info(default)|debug|trace

global:
  max:
    ue: 1024 # The number of UE can be increased depending on memory size.
    # peer: 64

nrf:
  serving: # 5G roaming requires PLMN in NRF
    - plmn_id:
        mcc: 999
        mnc: 70
    sbi:
      server:
        - address: 127.0.0.10
          port: 7777
```

amf.yaml

```
GNU nano 8.4 amf.yaml
logger:
  file:
    path: /home/open5gscp/open5gs-2.7.7/install/var/log/open5gs/amf.log
  # level: info # fatal|error|warn|info(default)|debug|trace

global:
  max:
    ue: 1024 # The number of UE can be increased depending on memory size.
    # peer: 64

amf:
  sbi:
    server:
      - address: 192.168.200.128
        port: 7779
    client:
      - nrf:
          - uri: http://127.0.0.10:7777
      scp:
          - uri: http://127.0.0.10:7777
    ngap:
      server:
        - address: 192.168.200.128
      metrics:
        server:
          - address: 192.168.200.128
            port: 9090
    guami:
      - plmn_id:
          mcc: 999
          mnc: 70
        amf_id:
          region: 2
          set: 1
    tai:
      - plmn_id:
          mcc: 999
          mnc: 70
        tac: 1
```

smf.yaml

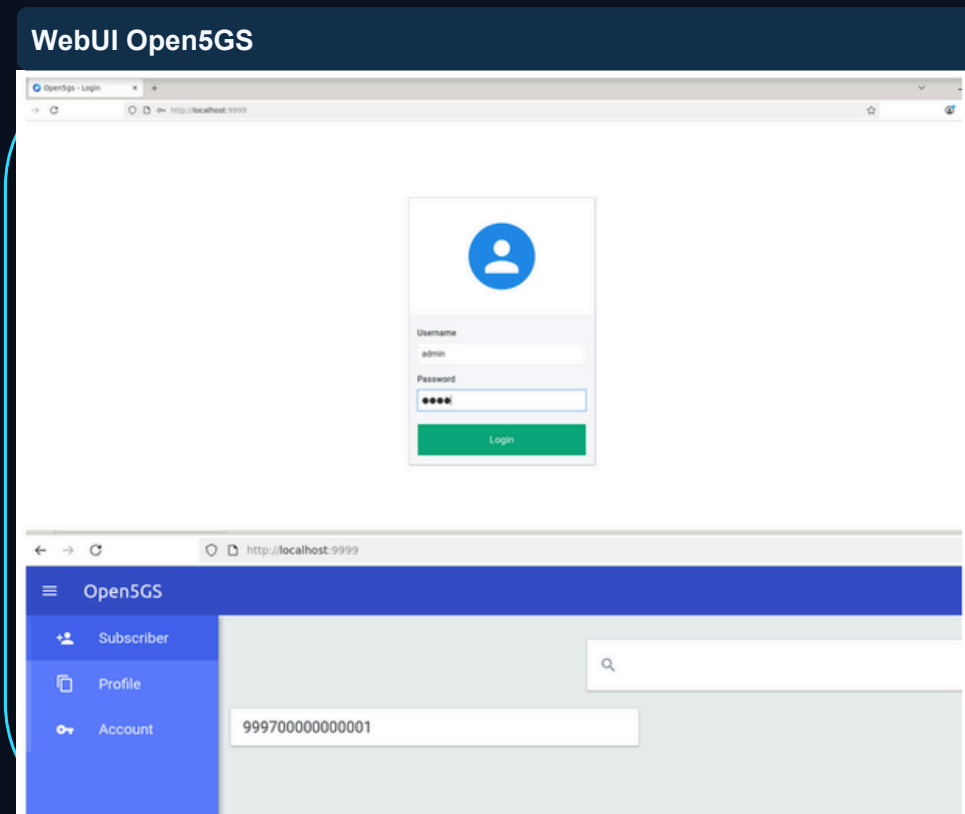
```
GNU nano 8.4 /home/open5gscp/open5gs-2.7.7/install/etc/open5gs/smf.yaml
file:
  path: /home/open5gscp/open5gs-2.7.7/install/var/log/open5gs/smf.log
  # level: info # fatal|error|warn|info(default)|debug|trace

global:
  max:
    ue: 1024 # The number of UE can be increased depending on memory size.
    # peer: 64

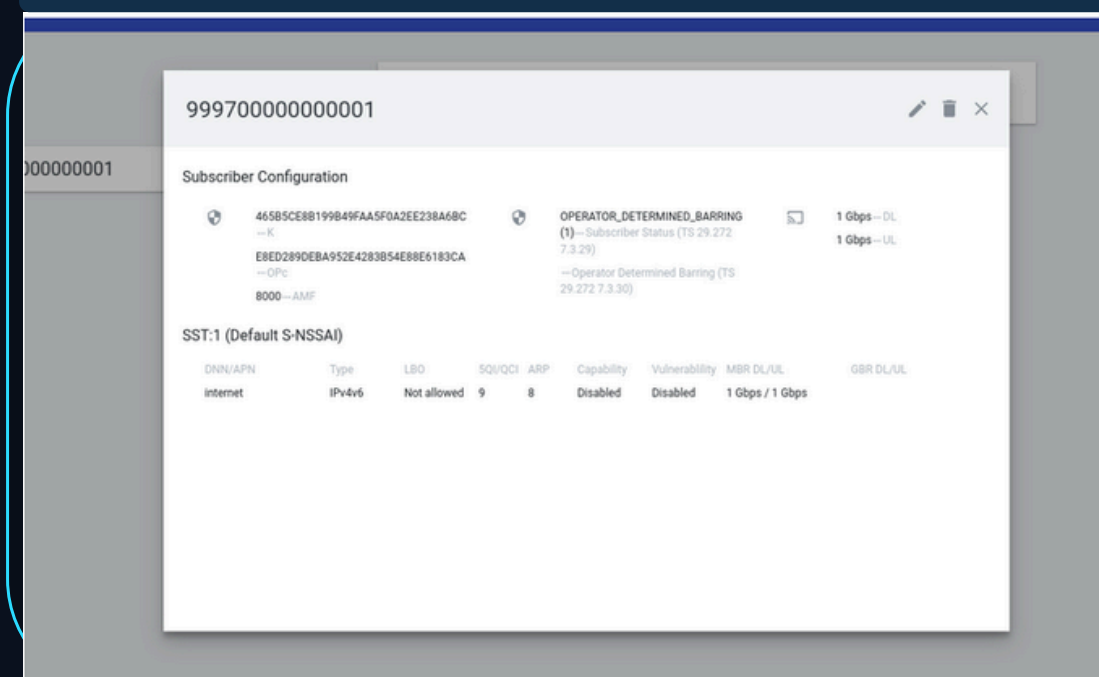
smf:
  sbi:
    server:
      - address: 192.168.200.128
        port: 7777
    client:
      - nrf:
          - uri: http://127.0.0.10:7777
      scp:
          - uri: http://192.168.200.128:7778
    pfcps:
      server:
        - address: 192.168.200.128
      client:
        upf:
          - address: 192.168.200.129
    gtpc:
      server:
        - address: 192.168.200.128
    gtpu:
      server:
        - address: 192.168.200.128
    metrics:
      server:
        - address: 192.168.200.128
          port: 9090
    sessions:
      - subnet: 10.45.0.0/16
        gateway: 10.45.0.1
      - subnet: 2001:db8:cafe::/48
        gateway: 2001:db8:cafe::1
```

Création et vérification de l'abonné

Le profil UE doit être cohérent entre UERANSIM, Open5GS et MongoDB



Profil abonné



MongoDB

```
test> use open5gs
switched to db open5gs
open5gs> db.subscribers.find().pretty()
{
  _id: ObjectId('6a29f769214618aac9df8a3'),
  imsi: '9997000000000001',
  msisdn: [],
  imeisv: [],
  mme_host: [],
  mme_realn: [],
  purge_flag: [],
  security: {
    k: '46585CE8B199B49FAA5F0A2EE238A6BC',
    op: null,
    opc: 'EBED289DEBA952E4283B54EB8E6183CA',
    amf: '8000'
  },
  ambr: { downlink: { value: 1, unit: 3 }, uplink: { value: 1, unit: 3 } },
  slice: {
    sst: 1,
    default_indicator: true,
    session: {
      name: 'internet',
      type: 3,
      qos: {
        index: 9,
        arp: {
          priority_level: 8,
          pre_emption_capability: 1,
          pre_emption_vulnerability: 1
        }
      }
    },
    ambr: { downlink: { value: 1, unit: 3 }, uplink: { value: 1, unit: 3 } }
  }
}
```

Éléments validés

IMSI/SUPI • clé d'authentification • OPc • DNN = internet • Slice SST = 1 • QoS

Configuration de l'UPF : plan utilisateur

L'UPF applique les règles reçues du SMF et transporte les données du UE

upf.yaml

```
GNU nano 8.4 upf.yaml *
logger:
  file:
    path: /home/open5gs/open5gs-2.7.7/install/var/log/open5gs/upf.log
  # level: info # fatal|error|warn|info(default)|debug|trace
global:
  max:
    ue: 1024 # The number of UE can be increased depending on memory size.
    # peer: 64
upf:
  pfcf:
    server:
      - address: 192.168.200.129
    client:
      # smf: # UPF PFCF Client try to associate SMF PFCF Server
      - address: 127.0.0.4
  gtpu:
    server:
      - address: 192.168.200.129
  session:
    - subnet: 10.45.0.0/16
    gateway: 10.45.0.1
    - subnet: 2001:db8:cafe::/48
    gateway: 2001:db8:cafe::1
  metrics:
    server:
      - address: 127.0.0.7
      port: 9090
#####
# PFCF Server
#####
# o Override PFCF address to be advertised to SMF in PFCF association
# pfcf:
#   server:
#     - dev: eth0
#     advertise: open5gs-upf.svc.local
#####
# GTP-U Server
#####
^G Aide ^O Écrire ^F Chercher ^K Couper ^T Exécuter ^C Emplacement M-U Ann
```

Ports PFCF / GTP-U et interface ogstun

```
open5gs@open5gs:~$ pgrep -a open5gs-upfd
3064 ./install/bin/open5gs-upfd -c ./install/etc/open5gs/upf.yaml
open5gs@open5gs:~$ sudo ss -ltnp | grep 8805
[sudo] Mot de passe de open5gs :
UNCONN 0 0 192.168.200.129:8805 0.0.0.0:* users:(("open5gs-upfd",pid=3064,fd=8))
open5gs@open5gs:~$ sudo ss -ltnp | grep 2152
UNCONN 0 0 192.168.200.129:2152 0.0.0.0:* users:(("open5gs-upfd",pid=3064,fd=9))
open5gs@open5gs:~$ ip -br address show ogstun
ogstun UNKNOWN 10.45.0.1/16 fe80::91f4:885a:621:5ca3/64
open5gs@open5gs:~$ ip route | grep 10.45
10.45.0.0/16 dev ogstun proto kernel scope link src 10.45.0.1
```

Configuration clé

PFCF : 192.168.200.129:8805
 GTP-U : 192.168.200.129:2152
 Réseau UE : 10.45.0.0/16
 Passerelle : 10.45.0.1

Validation du fonctionnement du réseau 5G

gNB : connexion SCTP et NG Setup

réussi et PFCP visible

```
ueransim@ueransim:~/opt/UERANSIM$ sudo ./build/nr-gnb -c config/open5gs-gnb.yaml
[sudo] Mot de passe de ueransim :
UERANSIM v3.2.8
[2026-06-16 19:37:36.533] [sctp] [info] Trying to establish SCTP connection... (192.168.200.128:38412)
[2026-06-16 19:37:36.543] [sctp] [info] SCTP connection established (192.168.200.128:38412)
[2026-06-16 19:37:36.544] [sctp] [debug] SCTP association setup ascId[4]
[2026-06-16 19:37:36.544] [ngap] [debug] Sending NG Setup Request
[2026-06-16 19:37:36.545] [ngap] [debug] NG Setup Response received
[2026-06-16 19:37:36.545] [ngap] [info] NG Setup procedure is successful
[2026-06-16 19:38:04.618] [rrc] [debug] UE[1] new signal detected
[2026-06-16 19:38:07.119] [rrc] [info] RRC Setup for UE[1]
[2026-06-16 19:38:07.123] [ngap] [debug] Initial NAS message received from UE[1]
[2026-06-16 19:38:07.219] [ngap] [debug] UE Context Release Command received
[2026-06-16 19:38:07.219] [rrc] [info] Releasing RRC connection for UE[1]
[2026-06-16 19:38:17.635] [rrc] [info] RRC Setup for UE[1]
```

UE : IP uesimtun0 et ping

```
ueransim@ueransim:~$ ip -br address show uesimtun0
uesimtun0 UNKNOWN 10.45.0.3/24 fe80::38e4:2a7d:f8d:15ee/64
ueransim@ueransim:~$ ip route show table rt_uesimtun0
default dev uesimtun0 scope link
ueransim@ueransim:~$ ping -I uesimtun0 -c 4 10.45.0.1
PING 10.45.0.1 (10.45.0.1) from 10.45.0.3 uesimtun0: 56(84) bytes of data.
64 bytes from 10.45.0.1: icmp_seq=1 ttl=64 time=5.15 ms
64 bytes from 10.45.0.1: icmp_seq=2 ttl=64 time=2.84 ms
^C
--- 10.45.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.843/3.996/5.150/1.153 ms
```

Ping et tcpdump PFCP

```
ueransim@ueransim:~$ ip -br address show uesimtun0
uesimtun0 UNKNOWN 10.45.0.8/24 fe80::7cc8:da52:bf2b:8e41/64
ueransim@ueransim:~$ ping -I uesimtun0 -c 4 10.45.0.1
PING 10.45.0.1 (10.45.0.1) from 10.45.0.8 uesimtun0: 56(84) bytes of data.
64 bytes from 10.45.0.1: icmp_seq=1 ttl=64 time=6.02 ms
64 bytes from 10.45.0.1: icmp_seq=2 ttl=64 time=4.25 ms
64 bytes from 10.45.0.1: icmp_seq=3 ttl=64 time=4.03 ms
64 bytes from 10.45.0.1: icmp_seq=4 ttl=64 time=3.35 ms

--- 10.45.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 3.346/4.410/6.024/0.988 ms

open5gs@open5gs:~$ sudo tcpdump -ni ens37 udp port 8805
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens37, link-type EN10MB (Ethernet), snapshot length 262144 bytes
15:51:42.263423 IP 192.168.200.129.8805 > 192.168.200.128.8805: UDP, length 16
15:51:42.264194 IP 192.168.200.128.8805 > 192.168.200.129.8805: UDP, length 16
15:51:42.264415 IP 192.168.200.128.8805 > 192.168.200.129.8805: UDP, length 16
15:51:42.264515 IP 192.168.200.129.8805 > 192.168.200.128.8805: UDP, length 16
^C
4 packets captured
4 packets received by filter
0 packets dropped by kernel
```

Services actifs côté Open5GS

L'AMF reçoit la connexion du gNB et gère l'enregistrement du UE.

```
open5gs@open5gs:~$ systemctl is-active mongod
active
open5gs@open5gs:~$ pgrep -af open5gs-
2434 /home/open5gs/open5gs-2.7.7/install/bin/open5gs-nrfd -c /home/open5gs/open5gs-2.7.7/install/etc/open5gs/nrf.yaml
2435 /home/open5gs/open5gs-2.7.7/install/bin/open5gs-scpd -c /home/open5gs/open5gs-2.7.7/install/etc/open5gs/scp.yaml
2456 /home/open5gs/open5gs-2.7.7/install/bin/open5gs-ausfd -c /home/open5gs/open5gs-2.7.7/install/etc/open5gs/ausf.yaml
2457 /home/open5gs/open5gs-2.7.7/install/bin/open5gs-udmd -c /home/open5gs/open5gs-2.7.7/install/etc/open5gs/udm.yaml
2458 /home/open5gs/open5gs-2.7.7/install/bin/open5gs-udrfd -c /home/open5gs/open5gs-2.7.7/install/etc/open5gs/udr.yaml
2459 /home/open5gs/open5gs-2.7.7/install/bin/open5gs-pcfd -c /home/open5gs/open5gs-2.7.7/install/etc/open5gs/pcf.yaml
2511 sudo ./install/bin/open5gs-nssf -c ./install/etc/open5gs/nssf.yaml
2514 sudo ./install/bin/open5gs-nssf -c ./install/etc/open5gs/nssf.yaml
2522 sudo ./install/bin/open5gs-smfd -c ./install/etc/open5gs/smf.yaml
2529 sudo ./install/bin/open5gs-smfd -c ./install/etc/open5gs/smf.yaml
2530 ./install/bin/open5gs-smfd -c ./install/etc/open5gs/smf.yaml
2563 tail -n 0 -F /home/open5gs/open5gs-2.7.7/install/var/log/open5gs/amf.log /home/open5gs/open5gs-2.7.7/install/var/log/
open5gs-2.7.7/install/var/log/open5gs/udm.log /home/open5gs/open5gs-2.7.7/install/var/log/open5gs/udr.log
2533 sudo ./install/bin/open5gs-amfd -c ./install/etc/open5gs/amf.yaml
2537 sudo ./install/bin/open5gs-amfd -c ./install/etc/open5gs/amf.yaml
2538 ./install/bin/open5gs-amfd -c ./install/etc/open5gs/amf.yaml
open5gs@open5gs:~$ sudo ss -ltnA sctp | grep 38412
[sudo] Mot de passe de open5gs :
LISTEN 0 5 192.168.200.128:38412 0.0.0.0:* users:((("open5gs-amfd",pid=5338,fd=8))
open5gs@open5gs:~$ sudo ss -ltnA | grep 8805
UNCONN 0 0 192.168.200.128:8805 0.0.0.0:* users:((("open5gs-smfd",pid=3530,fd=19))
```


PFCP : protocole critique entre SMF et UPF

Interface N4 — UDP 8805 — contrôle des sessions utilisateur

Wireshark : trafic PFCP normal

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.200.128	192.168.200.129	PFCP	60	PFCP Heartbeat Request
2	0.000291367	192.168.200.129	192.168.200.128	PFCP	58	PFCP Heartbeat Response
3	0.000386836	192.168.200.129	192.168.200.128	PFCP	58	PFCP Heartbeat Request
4	0.001150739	192.168.200.128	192.168.200.129	PFCP	60	PFCP Heartbeat Response
5	7.745508230	192.168.200.128	192.168.200.129	PFCP	60	PFCP Session Deletion Request
6	7.746234805	192.168.200.129	192.168.200.128	PFCP	175	PFCP Session Deletion Response
7	7.997565650	192.168.200.128	192.168.200.129	PFCP	638	PFCP Session Establishment Request
8	7.999621459	192.168.200.129	192.168.200.128	PFCP	158	PFCP Session Establishment Response
9	8.008817288	192.168.200.128	192.168.200.129	PFCP	117	PFCP Session Modification Request
10	8.010042200	192.168.200.129	192.168.200.128	PFCP	63	PFCP Session Modification Response
11	11.004989286	192.168.200.128	192.168.200.129	PFCP	60	PFCP Heartbeat Request
12	11.005386282	192.168.200.129	192.168.200.128	PFCP	58	PFCP Heartbeat Response
13	11.005528560	192.168.200.129	192.168.200.128	PFCP	58	PFCP Heartbeat Request
14	11.005966774	192.168.200.128	192.168.200.129	PFCP	60	PFCP Heartbeat Response
15	22.010434711	192.168.200.129	192.168.200.128	PFCP	58	PFCP Heartbeat Request
16	22.010916166	192.168.200.128	192.168.200.129	PFCP	60	PFCP Heartbeat Response
17	22.011000374	192.168.200.129	192.168.200.128	PFCP	58	PFCP Heartbeat Request
18	22.011100700	192.168.200.128	192.168.200.129	PFCP	60	PFCP Heartbeat Response

tcpdump : flux PFCP sur UDP 8805

```
open5gscp@open5gscp:~$ sudo tcpdump -ni any -nn -vv 'udp port 8805 and host 192.168.200.129'
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
01:12:52.741411 ens37 Out IP (tos 0x0, ttl 64, id 16166, offset 0, flags [DF], proto UDP (17), length 44)
    192.168.200.128.8805 > 192.168.200.129.8805: [bad udp cksum 0x127d -> 0x3956!] UDP, length 16
01:12:52.742274 ens37 In  IP (tos 0x0, ttl 64, id 18126, offset 0, flags [DF], proto UDP (17), length 44)
    192.168.200.129.8805 > 192.168.200.128.8805: [udp sum ok] UDP, length 16
01:12:52.742275 ens37 In  IP (tos 0x0, ttl 64, id 18127, offset 0, flags [DF], proto UDP (17), length 44)
    192.168.200.129.8805 > 192.168.200.128.8805: [udp sum ok] UDP, length 16
01:12:52.742488 ens37 Out IP (tos 0x0, ttl 64, id 16167, offset 0, flags [DF], proto UDP (17), length 44)
    192.168.200.128.8805 > 192.168.200.129.8805: [bad udp cksum 0x127d -> 0xb854!] UDP, length 16
01:12:53.200291 ens37 Out IP (tos 0x0, ttl 64, id 16228, offset 0, flags [DF], proto UDP (17), length 44)
    192.168.200.128.8805 > 192.168.200.129.8805: [bad udp cksum 0x127d -> 0xa7e6!] UDP, length 16
01:12:53.200981 ens37 In  IP (tos 0x0, ttl 64, id 18169, offset 0, flags [DF], proto UDP (17), length 161)
    192.168.200.129.8805 > 192.168.200.128.8805: [udp sum ok] UDP, length 133
01:12:53.450884 ens37 Out IP (tos 0x0, ttl 64, id 16245, offset 0, flags [DF], proto UDP (17), length 624)
    192.168.200.128.8805 > 192.168.200.129.8805: [bad udp cksum 0x14c1 -> 0x7742!] UDP, length 596
01:12:53.452201 ens37 In  IP (tos 0x0, ttl 64, id 18187, offset 0, flags [DF], proto UDP (17), length 144)
    192.168.200.129.8805 > 192.168.200.128.8805: [udp sum ok] UDP, length 116
01:12:53.459350 ens37 Out IP (tos 0x0, ttl 64, id 16247, offset 0, flags [DF], proto UDP (17), length 103)
    192.168.200.128.8805 > 192.168.200.129.8805: [bad udp cksum 0x12b8 -> 0x3a96!] UDP, length 75
01:12:53.460452 ens37 In  IP (tos 0x0, ttl 64, id 18189, offset 0, flags [DF], proto UDP (17), length 49)
    192.168.200.129.8805 > 192.168.200.128.8805: [udp sum ok] UDP, length 21
01:13:03.745035 ens37 In  IP (tos 0x0, ttl 64, id 20387, offset 0, flags [DF], proto UDP (17), length 44)
    192.168.200.129.8805 > 192.168.200.128.8805: [udp sum ok] UDP, length 16
01:13:03.745073 ens37 Out IP (tos 0x0, ttl 64, id 18358, offset 0, flags [DF], proto UDP (17), length 44)
    192.168.200.128.8805 > 192.168.200.129.8805: [bad udp cksum 0x127d -> 0x3556!] UDP, length 16
01:13:03.745197 ens37 Out IP (tos 0x0, ttl 64, id 18359, offset 0, flags [DF], proto UDP (17), length 44)
    192.168.200.128.8805 > 192.168.200.129.8805: [bad udp cksum 0x127d -> 0xb754!] UDP, length 16
01:13:03.745725 ens37 In  IP (tos 0x0, ttl 64, id 20388, offset 0, flags [DF], proto UDP (17), length 44)
    192.168.200.129.8805 > 192.168.200.128.8805: [udp sum ok] UDP, length 16
```

Ce que PFCP permet

- Créer une session PDU
- Modifier une session existante
- Supprimer une session
- Installer les règles PDR/FAR/QR/URR
- Synchroniser le SMF et l'UPF

Analyse d'un échange PFCP normal

Session Establishment Request → Response → Modification

Session Establishment Request

```
Wireshark - Paquet 7 - ens37 (udp port 8805)

Destination Address: 192.168.200.129
[Stream index: 0]
> User Datagram Protocol, Src Port: 8805, Dst Port: 8805
  Packet Forwarding Control Protocol
    Flags: 0x21, SEID (S)
      001. .... = Version: 1
      ...0 .... = Spare: 0
      ....0... = Spare: 0
      ....0... = Follow On (FO): False
      ....0... = Message Priority (MP): False
      ....1... = SEID (S): True
    Message Type: PFCP Session Establishment Request (50)
    Length: 592
    SEID: 0x0000000000000000
    Sequence Number: 2598
    Spare: 0
    > Node ID : IPv4 address: 192.168.200.128
    > F-SEID : SEID: 0x00000000000000377, IPv4 192.168.200.128
    > Create PDR : [Grouped IE]: PDR ID: 1
      IE Type: Create PDR (1)
      IE Length: 74
      > PDR ID : 1
      > Precedence : 65535
      > PDI : [Grouped IE]
      > FAR ID : Dynamic by CP 1
      > URR ID : Dynamic by CP 1
      > QER ID : Dynamic by CP 1
    > Create PDR : [Grouped IE]: PDR ID: 2
    > Create PDR : [Grouped IE]: PDR ID: 3
      IE Type: Create PDR (1)
      IE Length: 41
      > PDR ID : 3
      > Precedence : 255
      > PDI : [Grouped IE]
      > Outer Header Removal : GTP-U/UDP/IP
      > FAR ID : Dynamic by CP 1
    > Create PDR : [Grouped IE]: PDR ID: 4
    > Create FAR : [Grouped IE]: FAR ID: Dynamic by CP 1
      IE Type: Create FAR (3)
      IE Length: 19
    0010 02 70 db ed 40 00 40 11 4a 3c c0 a8 c8 80 c0 a8 .p..@.@.Jc....[
    0020 c8 81 22 65 22 65 02 5c 2d 88 21 32 02 50 00 00 [e"e\..12.P..
    0030 00 00 00 00 00 00 00 0a 26 00 00 3c 00 05 00 c0 .....&..<...
    0040 a8 c8 80 00 39 00 0d 02 00 00 00 00 00 03 77 ....9....w
    0050 c0 a8 c8 80 00 01 00 4a 00 38 00 02 00 01 00 1d .....J.8.....
    0060 00 04 00 00 ff ff 00 02 00 20 00 14 00 01 01 00 ....inte rnet:]
    0070 16 00 09 08 69 6e 74 65 72 6e 65 74 00 5d 00 05 .....m.....
    0080 06 0a 2d 00 07 00 a0 00 01 11 00 6c 00 04 00 00 .....Q.....m...
    0090 00 01 00 51 00 04 00 00 00 01 00 6d 00 04 00 00 .....I.B.....
    00a0 00 01 00 01 00 49 00 38 00 02 00 02 00 1d 00 04 .....

No.: 7 - Time: 7.997565650 - Source: 192.168.200.128 - Destination: 192.168.200.129 - Protocol: PFCP - Length: 638 - Info: PFCP Session Establishment Request
```

Session Establishment Response

```
Wireshark - Paquet 8 - ens37 (udp port 8805)

> Frame 8: 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface ens37, id 0
  Ethernet II, Src: VMware_8c:d4:b3 (00:0c:29:8c:d4:b3), Dst: VMware_7b:f4:3a (00:0c:29:7b:f4:3a)
  Internet Protocol Version 4, Src: 192.168.200.129, Dst: 192.168.200.128
  User Datagram Protocol, Src Port: 8805, Dst Port: 8805
    Packet Forwarding Control Protocol
      Flags: 0x21, SEID (S)
        001. .... = Version: 1
        ...0 .... = Spare: 0
        ....0... = Spare: 0
        ....0... = Follow On (FO): False
        ....0... = Message Priority (MP): False
        ....1... = SEID (S): True
      Message Type: PFCP Session Establishment Response (51)
      Length: 112
      SEID: 0x00000000000000377
      Sequence Number: 2598
      Spare: 0
      > Node ID : IPv4 address: 192.168.200.129
      > Cause : Request accepted(success)
        IE Type: Cause (19)
        IE Length: 1
        Cause: Request accepted(success) (1)
      > F-SEID : SEID: 0x000000000000008f8, IPv4 192.168.200.129
      > Created PDR : [Grouped IE]: PDR ID: 2
      > Created PDR : [Grouped IE]: PDR ID: 3
      > Created PDR : [Grouped IE]: PDR ID: 4
      [Response To: 7]
      [Response Time: 0.002055809 seconds]
```

Session Modification Request

```
Wireshark - Paquet 9 - ens37 (udp port 8805)

> Frame 9: 117 bytes on wire (936 bits), 117 bytes captured (936 bits) on interface ens37, id 0
  Ethernet II, Src: VMware_7b:f4:3a (00:0c:29:7b:f4:3a), Dst: VMware_8c:d4:b3 (00:0c:29:8c:d4:b3)
  Internet Protocol Version 4, Src: 192.168.200.128, Dst: 192.168.200.129
  User Datagram Protocol, Src Port: 8805, Dst Port: 8805
    Packet Forwarding Control Protocol
      Flags: 0x21, SEID (S)
        001. .... = Version: 1
        ...0 .... = Spare: 0
        ....0... = Spare: 0
        ....0... = Follow On (FO): False
        ....0... = Message Priority (MP): False
        ....1... = SEID (S): True
      Message Type: PFCP Session Modification Request (52)
      Length: 71
      SEID: 0x000000000000008f8
      Sequence Number: 2599
      Spare: 0
      > Update FAR : [Grouped IE]: FAR ID: Dynamic by CP 1
        IE Type: Update FAR (10)
        IE Length: 55
        > FAR ID : Dynamic by CP 1
        > Apply Action :
        > Update Forwarding Parameters : [Grouped IE]
          IE Type: Update Forwarding Parameters (11)
          IE Length: 37
          > Destination Interface : Access
          > Network Instance : Internet
          > Outer Header Creation :
            > 3GPP Interface Type : N3 3GPP Access:
              IE Type: 3GPP Interface Type (160)
              IE Length: 1
              0000 .... = Spare: 0
              ..00 1011 = 3GPP Interface Type: N3 3GPP Access (11)
          [Response In: 10]
          0000 00 0c 29 8c d4 b3 00 0c 29 7b f4 3a 00 00 45 00 ..)...)(::E
          0010 00 07 db ed 40 00 40 11 4c 44 c0 a8 c8 80 c0 a8 .g.@.@.LD....
          0020 c8 81 22 65 22 65 00 53 e8 82 21 34 00 47 00 00 .."e"S..14.G..
          0030 00 00 00 00 08 f8 00 0a 27 00 00 0a 00 37 00 6c .....7..l
          0040 00 04 00 00 01 00 2c 00 02 02 00 00 00 00 25 .....%
          0050 00 01 00 00 16 00 09 08 69 6e 74 65 72 6e [.....Intern
          0060 65 74 00 54 00 0a 01 00 00 00 00 06 c0 a8 c8 82 et T.....
          0070 00 a0 00 01 0b .....
```

Le SMF programme l'UPF : création des règles PDR, FAR, QER et URR, puis adaptation de la session si nécessaire.

Analyse de l'attaque PFCP

Observation d'un trafic répétitif et anormal de Heartbeat Request

Wireshark : attaque PFCP Heartbeat Flood

Partie attaque protocole PFCP

No.	Time	Source	Destination	Protocol	Length	Info
147	13.685500884	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
148	13.733454800	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
149	13.773608194	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
150	13.818149372	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
151	13.862385750	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
152	13.910299340	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
153	13.961817093	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
154	14.001281195	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
155	14.057569073	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
156	14.101710122	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
157	14.154065696	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
158	14.217706453	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
159	14.257992678	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
160	14.305424993	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
161	14.345790102	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
162	14.390288433	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
163	14.442382235	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
164	14.481869034	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
165	14.525446316	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
166	14.569859417	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
167	14.614565729	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request
168	14.661675176	192.168.200.150	192.168.200.129	PFCP	60	PFCP Heartbeat Request

Frame 209: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface ens37, id 0

Ethernet II, Src: VMware_91:bd:0b (00:0c:29:91:bd:0b), Dst: VMware_8c:d4:b3 (00:0c:29:8c:d4:b3)

- Destination: VMware_8c:d4:b3 (00:0c:29:8c:d4:b3)
- Source: VMware_91:bd:0b (00:0c:29:91:bd:0b)
- Type: IPv4 (0x0800)
- [Stream index: 1]
- Padding: 0000

Internet Protocol Version 4, Src: 192.168.200.150, Dst: 192.168.200.129

User Datagram Protocol, Src Port: 8805, Dst Port: 8805

- Source Port: 8805
- Destination Port: 8805
- Length: 24
- Checksum: 0xa5e8 [unverified]
- [Checksum Status: Unverified]
- [Stream index: 1]
- [Stream Packet Number: 201]
- [Timestamps]
- UDP payload (16 bytes)

Packet Forwarding Control Protocol

- Flags: 0x20
- Message Type: PFCP Heartbeat Request (1)
- Length: 12

Observation

Nombreux Heartbeat Request
Source externe vers UPF
Destination : UDP 8805
Répétition anormale
Risque : surcharge de l'interface N4

Impacts possibles d'une attaque PFCP

Une attaque sur N4 peut dégrader la session data sans casser la connexion radio

Heartbeat Flood

Surcharge de traitement du UPF

Session Flood

Création excessive de sessions

Session Deletion

Coupure du trafic utilisateur

Session Modification

Débit instable / QoS dégradée

Malformed PFCP

Erreurs de parsing ou crash

Effet utilisateur

Connecté mais Internet instable

Point clé : les phases RRC/NAS peuvent fonctionner, alors que la phase session/data est perturbée.

Recommandations de sécurité

Réduire l'exposition de l'interface N4 et détecter les anomalies PFCP

Mesures réseau

- Limitier l'accès à N4 aux IP autorisées
- Filtrer le port UDP 8805
- Superviser le volume de messages PFCP
- Journaliser les événements SMF/UPF

Détection et supervision

- Détecter les messages anormaux
- Mettre du rate limiting
- Surveiller SEID et types de messages
- Envisager IPsec/TLS selon le contexte

Conclusion

Un réseau 5G simulé fonctionnel et une zone critique clairement identifiée

Ce qui a été réalisé

- Déploiement UERANSIM + Open5GS
- Configuration UE, gNB, CP et UPF
- Création et validation d'une session PDU
- Observation des échanges PFCP
- Analyse d'une attaque Heartbeat Flood

Message final

PFCP est discret mais critique : sa compromission peut impacter directement la connectivité des utilisateurs.

Il doit donc être filtré, supervisé et surveillé spécifiquement.